

# Seminar 0: Review of Complex Numbers

## Questions and Solutions

### 1. The imaginary unit and quadratic equations

Recall that  $i^2 = -1$ .

Solve over the complex numbers.

**(a)**  $x^2 + 9 = 0$

**Solution:**  $x = \pm 3i$ .

**(b)**  $x^2 - 2x + 5 = 0$

**Solution:**  $x = 1 \pm 2i$ .

**(c)**  $5x^2 - 2x + 1 = 0$

**Solution:**  $x = \frac{1 \pm 2i}{5}$ .

**(d)**  $x^3 + 2x^2 + x = 0$

**Solution:**  $x = 0$  or  $x = -1$ , with  $x = -1$  a double root.

### 2. Real and imaginary parts

For each complex number, identify its real and imaginary parts.

**(a)**  $z_1 = 1 + i$

**Solution:**  $\operatorname{Re}(z_1) = 1$ ,  $\operatorname{Im}(z_1) = 1$ .

**(b)**  $z_2 = 3 - 2i$

**Solution:**  $\operatorname{Re}(z_2) = 3$ ,  $\operatorname{Im}(z_2) = -2$ .

**(c)**  $z_3 = -4i$

**Solution:**  $\operatorname{Re}(z_3) = 0$ ,  $\operatorname{Im}(z_3) = -4$ .

**(d)**  $z_4 = 7$

**Solution:**  $\operatorname{Re}(z_4) = 7$ ,  $\operatorname{Im}(z_4) = 0$ .

**(e)**  $z_5 = -2 + 5i$

**Solution:**  $\operatorname{Re}(z_5) = -2$ ,  $\operatorname{Im}(z_5) = 5$ .

### 3. Addition, subtraction, and multiplication

Let  $z_1 = 1 + i$ ,  $z_2 = 3 - 2i$ .

Compute in Cartesian form.

**(a)**  $z_1 + z_2$

**Solution:**  $4 - i$ .

**(b)**  $z_1 - z_2$

**Solution:**  $-2 + 3i$ .

**(c)**  $z_2 - z_1$

**Solution:**  $2 - 3i$ .

**(d)**  $z_1 z_2$

**Solution:**  $5 + i$ .

**(e)**  $z_1^2$

**Solution:**  $2i$ .

**(f)**  $z_2^2$

**Solution:**  $5 - 12i$ .

#### 4. Division of complex numbers

Compute in Cartesian form.

**(a)**  $\frac{1+i}{3-2i}$

**Solution:**  $\frac{1}{13} + \frac{5}{13}i$ .

**(b)**  $\frac{2-i}{1+i}$

**Solution:**  $\frac{1}{2} - \frac{3}{2}i$ .

**(c)**  $\frac{1}{2+3i}$

**Solution:**  $\frac{2}{13} - \frac{3}{13}i$ .

**(d)**  $\frac{4i}{1-i}$

**Solution:**  $-2 + 2i$ .

**(e)** Solve  $(3 - 2i)z = 1 + i$ .

**Solution:**  $z = \frac{1}{13} + \frac{5}{13}i$ .

#### 5. Powers of the imaginary unit

Compute.

**(a)**  $i^0, i^1, i^2, i^3, i^4$

**Solution:**  $1, i, -1, -i, 1$ .

**(b)**  $i^{17}$

**Solution:**  $i$ .

**(c)**  $i^{38}$

**Solution:**  $-1$ .

**(d)**  $i^{2001}$

**Solution:**  $i$ .

**(e)**  $(-i)^{23}$

**Solution:**  $i$ .

**(f)** State the general rule for  $i^n$ ,  $n \geq 0$ .

**Solution:** If  $n = 4q + r$ , where  $r \in \{0, 1, 2, 3\}$ , then  $i^n = i^r$ .

#### 6. Complex conjugation

Compute the conjugates.

**(a)**  $1 + i$

**Solution:**  $1 - i$ .

**(b)**  $3 - 2i$

**Solution:**  $3 + 2i$ .

(c)  $-4i$

**Solution:**  $4i$ .

(d)  $5$

**Solution:**  $5$ .

(e)  $\sqrt{3} + 2i$

**Solution:**  $\sqrt{3} - 2i$ .

(f) Let  $z = x + iy$ . Compute  $z\bar{z}$ .

**Solution:**  $z\bar{z} = x^2 + y^2$ .

(g) Verify the conjugate identities.

**Solution:**  $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$ ,  $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$ .

## 7. Recovering real and imaginary parts

Let  $z = x + iy$ .

(a) Express  $\operatorname{Re}(z)$  using  $z$  and  $\bar{z}$ .

**Solution:**  $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$ .

(b) Express  $\operatorname{Im}(z)$  using  $z$  and  $\bar{z}$ .

**Solution:**  $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$ .

(c) Apply the formulas to  $z = 7 - 5i$ .

**Solution:**  $\operatorname{Re}(z) = 7$ ,  $\operatorname{Im}(z) = -5$ .

## 8. Modulus and distance

Compute the modulus.

(a)  $|1 + i|$

**Solution:**  $\sqrt{2}$ .

(b)  $|3 - 4i|$

**Solution:**  $5$ .

(c)  $|-5i|$

**Solution:**  $5$ .

(d)  $|-2 + 2i|$

**Solution:**  $2\sqrt{2}$ .

(e)  $|\sqrt{3} + i|$

**Solution:**  $|3 + i| = \sqrt{3^2 + 1^2} = \boxed{\sqrt{10}}$ .

For  $z = x + iy$ , express the following in terms of  $x$  and  $y$ .

(f)  $|z + 7i|$

**Solution:**  $\sqrt{x^2 + (y + 7)^2}$ .

(g)  $|z - 2|$

**Solution:**  $\sqrt{(x - 2)^2 + y^2}$ .

(h)  $|z - (5 + i)|$

**Solution:**  $\sqrt{(x - 5)^2 + (y - 1)^2}$ .

## 9. Cartesian and polar forms

Write each number as  $z = re^{i\theta}$ , using the stated principal argument.

(a)  $1 + i$

**Solution:**  $\sqrt{2}e^{i\pi/4}$ .

(b)  $3 - 2i$

**Solution:**  $\sqrt{13}e^{-i\arctan(2/3)}$ .

(c)  $1 + \sqrt{3}i$

**Solution:**  $1 + 3i = \sqrt{10}e^{i\arctan 3}$ .

(d)  $-1 + i$

**Solution:**  $\sqrt{2}e^{i3\pi/4}$ .

(e)  $-2 - 2i$

**Solution:**  $2\sqrt{2}e^{-i3\pi/4}$ .

(f)  $\frac{1+i}{1-i}$

**Solution:**  $i = e^{i\pi/2}$ .

## 10. Euler's formula

(a) State Euler's formula.

**Solution:**  $e^{i\theta} = \cos\theta + i\sin\theta$ .

(b) Express  $\cos\theta$  using complex exponentials.

**Solution:**  $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ .

(c) Express  $\sin\theta$  using complex exponentials.

**Solution:**  $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$ .

## 11. Multiplication in polar form

Let  $z_1 = r_1e^{i\theta_1}$ ,  $z_2 = r_2e^{i\theta_2}$ .

(a) Compute  $z_1z_2$ .

**Solution:**  $z_1z_2 = r_1r_2e^{i(\theta_1+\theta_2)}$ .

(b) State the modulus and argument rules.

**Solution:**  $|z_1z_2| = |z_1||z_2|$ ,  $\arg(z_1z_2) = \arg(z_1) + \arg(z_2) \pmod{2\pi}$ .

(c) Verify for  $z_1 = 1 + i$  and  $z_2 = \sqrt{3} + i$ .

**Solution:**  $z_1z_2 = (\sqrt{3} - 1) + (\sqrt{3} + 1)i = 2\sqrt{2}e^{i7\pi/12}$ .

## 12. De Moivre's formula

(a) State De Moivre's formula.

**Solution:**  $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$ .

(b) Compute  $(1 + i)^8$ .

**Solution:**  $16 = 16e^{i0}$ .

(c) Compute  $(\sqrt{3} + i)^6$ .

**Solution:**  $(3 + i)^6 = -352 + 936i$ .

(d) Compute  $(1 - i)^{10}$ .

**Solution:**  $(1 - i)^{10} = -32i$ .

### 13. Roots of complex numbers

Find all roots.

(a)  $z^3 = 1$

**Solution:**  $z_k = e^{i2\pi k/3}, \quad k = 0, 1, 2.$

(b)  $z^4 = 1$

**Solution:**  $z_k = e^{i\pi k/2}, \quad k = 0, 1, 2, 3.$

(c)  $z^4 = -1$

**Solution:**  $z_k = e^{i(\pi/4 + k\pi/2)}, \quad k = 0, 1, 2, 3.$

(d)  $z^6 = 64$

**Solution:**  $z_k = 2e^{i\pi k/3}, \quad k = 0, 1, 2, 3, 4, 5.$

### 14. Geometric interpretation

Describe each transformation and apply it to  $z = 1 + i$ .

(a)  $z \mapsto iz$

**Solution:** Rotation by  $\pi/2$ ;  $1 + i \mapsto -1 + i$ .

(b)  $z \mapsto -z$

**Solution:** Rotation by  $\pi$ ;  $1 + i \mapsto -1 - i$ .

(c)  $z \mapsto \bar{z}$

**Solution:** Reflection in the real axis;  $1 + i \mapsto 1 - i$ .

(d)  $z \mapsto 2z$

**Solution:** Scaling by 2;  $1 + i \mapsto 2 + 2i$ .

(e)  $z \mapsto e^{i\pi/4}z$

**Solution:** Rotation by  $\pi/4$ ;  $1 + i \mapsto i\sqrt{2}$ .

(f)  $z \mapsto 2e^{i\pi/3}z$

**Solution:** Scaling by 2 and rotation by  $\pi/3$ ;  $1 + i \mapsto (1 - \sqrt{3}) + (1 + \sqrt{3})i$ .

### 15. Basic identities and inequalities

State the results.

(a)  $|z|^2$

**Solution:**  $|z|^2 = z\bar{z}$ .

(b)  $|\bar{z}|$

**Solution:**  $|\bar{z}| = |z|$ .

(c)  $|zw|$

**Solution:**  $|zw| = |z| |w|$ .

(d)  $\left| \frac{z}{w} \right|, w \neq 0$

**Solution:**  $\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$ .

(e) Triangle inequality

**Solution:**  $|z + w| \leq |z| + |w|$ .

(f) Distance between the points represented by  $z$  and  $w$

**Solution:**  $|z - w|$ .